

Ego-Human Motion Prediction with 3D-Aware LLM

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Abstract. Anticipating human motion from an egocentric perspective is fundamental for proactive assistance in AR/VR, human-robot collaboration, and embodied AI. While recent works incorporate language as a semantic prior to reduce the ill-posed nature of egocentric forecasting, they largely neglect the 3D spatial and semantic context that governs how motion unfolds, and treat pose and language prediction as separate inference streams. We introduce Ego3DLM, built on two core principles: accurate motion forecasting requires explicit spatial and semantic understanding of the 3D environment, and pose and language must be predicted holistically in a single pass, since motion is inherently tied to the semantic interpretation of actions being performed. Given three-point tracking, 3D scene features, and egocentric video, Ego3DLM simultaneously decodes past pose, future pose, past narration, and future narration in a single autoregressive pass, grounding predicted poses and descriptions in one another to enforce cross-modal and temporal consistency. We adopt a three-stage training scheme: (1) spatial-semantic scene awareness pretraining; (2) holistic instruction tuning over all four outputs in a single pass; and (3) GRPO-based reinforcement finetuning with intra- and inter-modal rewards that directly optimize pose-language fidelity. Experiments on the Nymeria benchmark demonstrate that Ego3DLM achieves state-of-the-art performance across future motion prediction, past motion tracking, and motion description, showing that 3D scene grounding and holistic cross-modal prediction yield physically plausible and semantically coherent motion forecasts. The project page is available at <https://jaewoo97.github.io/Ego3DLM/>.

Keywords: Egocentric Vision · Human Motion Forecasting · Multi-modal Large Language Models

1 Introduction

Predicting human motion from an egocentric perspective is fundamental for proactive, real-time assistance in AR/VR, human-robot collaboration, and embodied AI [26, 46, 47, 53, 69, 72, 73, 75]. Unlike third-person footage, egocentric observations reflect the wearer’s true field of view and thus encode intent, affordances, and near-field interactions that matter for planning [21, 22, 39]. Yet

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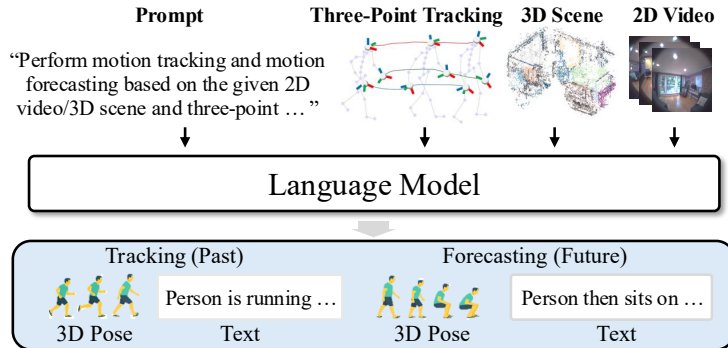


Fig. 1: Our Ego3DLM incorporates the semantic context of the surrounding 3D environment along with 2D egocentric video and three-point tracking data to generate past and future poses and their corresponding language motion descriptions.

forecasting from this viewpoint is inherently ill-posed: the camera captures little of the body, only sparse wearable cues are available, and the same partial observation is consistent with many plausible future motions. Resolving this ambiguity requires grounding predictions in semantic context: understanding not merely the kinematic state of the body, but the underlying intent and action semantics that govern how motion will unfold.

Towards this end, recent works have sought to reduce forecast ambiguity by incorporating language as a semantic prior on human intention, demonstrating that treating motion as a tokenized sequence enables unified reasoning across vision, language, and pose [30, 36]. Motion-language models such as MotionGPT [36] and EgoLM [30] cast 3D human motion as “motion tokens” for joint modeling with text, while pose-language systems such as ChatPose [20] and PoseScript [15] demonstrate that large language models can understand and generate fine-grained 3D human poses from textual descriptions. However, most existing methods emphasize reconstruction, captioning, or editing of observed motion rather than prediction, and often operate on third-person videos or canonicalized motion capture rather than egocentric inputs.

Forecasting works on egocentric motion, such as EgoCast [17] and UniEgoMotion [52], focus on pose prediction from egocentric videos and sparse proprioceptive signals. However, these methods typically omit explicit 3D scene representations whose spatial and semantic context is crucial for resolving the ambiguity of future motion in cluttered, contact-rich environments. While Fiction [1] and HMD² [27] employ an explicit 3D scene representation for pose prediction and motion generation, they inject these features directly into the model, which does not guarantee that the model acquires genuine 3D reasoning capabilities. Furthermore, prior works treat pose and language as separate inference streams, forgoing the cross-modal consistency that arises from holistic predictions and missing the opportunity to leverage semantic reasoning as a direct inductive bias during forecasting.

We address these limitations with Ego3DLM, built on two core principles. First, accurate human motion prediction requires explicit spatial and semantic understanding of the 3D environment, which defines the physical constraints, free space, and object-level affordances that govern possible movements. Second, pose and language must be predicted holistically in a single pass rather than independently, since human motion is inherently tied to the semantic interpretation of the actions being performed. To this end, our model takes three-point tracking (*i.e.* head and hand poses), 3D scene features, and egocentric video as input, and simultaneously decodes past pose, future pose, past text, and future text in a single forward pass (Fig. 1), grounding predicted poses and descriptions in one another to improve cross-modal and temporal consistency.

These principles directly motivate our three-stage training scheme. In Stage I, we ground the language model (LM) in the 3D environment by pretraining on automatically generated spatial and semantic question-answer (QA) pairs from 3D scenes, instilling obstacle awareness and object-level semantics before introducing motion reasoning. In Stage II, the scene-aware LM is instruction-tuned to holistically generate all four outputs in a single autoregressive pass, with an explicit spatial scene reasoning step prepended so that spatial understanding propagates through pose prediction and into language narration. In Stage III, Group Relative Policy Optimization (GRPO) with intra- and inter-modal rewards directly optimizes for pose and language accuracy and their fidelity, which likelihood-based training alone cannot enforce. This stage fully leverages cross-modal coherence between pose and language to encourage the model toward predictions that are both physically plausible and semantically consistent.

We validate Ego3DLM on the Nymeria [45] egocentric motion benchmark, which provides synchronized egocentric videos, language descriptions, and reconstructed 3D scenes. Our model achieves state-of-the-art performance across all metrics for future motion prediction and past motion tracking, along with their corresponding language descriptions, demonstrating comprehensive cross-modal understanding of human motion and intent. Our contributions are three-fold:

- **3D scene-aware egocentric motion prediction:** a unified framework that jointly forecasts language and 3D pose for both past and future from three-point cues, egocentric video, and explicit 3D scene features.
- **Simultaneous four-output decoding:** a single-pass generation of past pose, past text, future pose, and future text that enforces temporal and cross-modal coherence, improving fidelity between predicted motion and language.
- **Multi-stage training:** a three-stage training scheme that progressively builds from 3D spatial-semantic scene understanding to holistic egocentric motion tracking and forecasting with joint pose-language supervision.

2 Related Works

2.1 Multi-Modal Language Models

Recent large language models (LLMs) have expanded beyond text and RGB images to operate on diverse modalities such as event streams [43, 56, 70, 81, 82],

point clouds [7, 10, 14, 23, 31, 32, 38, 74, 80], and 3D human pose [8, 18, 20, 36, 64, 79, 84]. EventGPT [43] extends LLMs to event cameras via spatio-temporal aggregation, while 3D-oriented models align point cloud encoders with language models [10, 23, 31]. Other approaches [30, 36, 84] tokenize SMPL [44] parameters as pose tokens [33, 66], and EgoLM [30] models the joint distribution of egocentric motion and language for wearable settings. Moreover, discrete motion tokens have been treated as a foreign “language”, enabling unified motion-language reasoning [36]. While these works show that heterogeneous inputs (events, points, pose) can be projected into a shared token space, most focus on understanding or reconstruction, with limited attention to forecasting multi-modal futures from explicit 3D scene geometry. Building upon them, we incorporate 3D scene-conditioned forecasting to predict human motion guided by semantic objectives.

2.2 Human Motion Understanding & Forecasting

Text-motion models [2, 24, 25, 54, 55, 65, 71, 76–78] have explored bidirectional mappings between natural language and 3D human motion. TEACH [2] composes long motions from paragraphs by stitching atomic actions, while diffusion-based models such as MDM [65] and MotionDiffuse [77] generate realistic and diverse motions from text by viewing motion synthesis as a denoising process in pose space. MotionGPT [36] unifies multiple motion-language tasks (captioning, text-driven generation, prediction) under a single LLM-style architecture by discretizing motion into tokens. These approaches show that language supervision can ground high-level semantics of human motion, but they typically operate in third-person views. EgoLM [30] extends motion-language modeling to the egocentric setting by jointly leveraging egocentric video and three-point tracking data. Nevertheless, these works do not explicitly account for the surrounding 3D scene, limiting their ability to anticipate scene-dependent contacts and physical constraints.

Human motion forecasting targets future 3D body pose [6, 9, 13, 35, 46, 48, 60, 62, 68, 75, 83], hand pose, or trajectory [3, 4, 12, 34, 37, 49, 51, 57]. In the egocentric setting, EgoCast [17] and UniEgoMotion [52] forecast 3D body pose, while hand-centric models such as HOI-Forecast [42] and EgoH4 [28] predict 3D hand motion and contact hotspots. Yet these methods typically predict either pose or hand trajectories alone and lack rich language descriptions. A separate line of work, F1ction [1] and HMD² [27], grounds motion prediction and tracking in 3D scenes, but injects scene geometry directly and encodes it implicitly, without explicit scene reasoning. In contrast, our model jointly predicts 3D pose and textual descriptions of both past and future from egocentric video and sparse three-point tracking, while reasoning explicitly over 3D scene geometry. This encourages diverse yet physically and semantically coherent motion-language futures that respect scene layout, obstacles, and affordances.

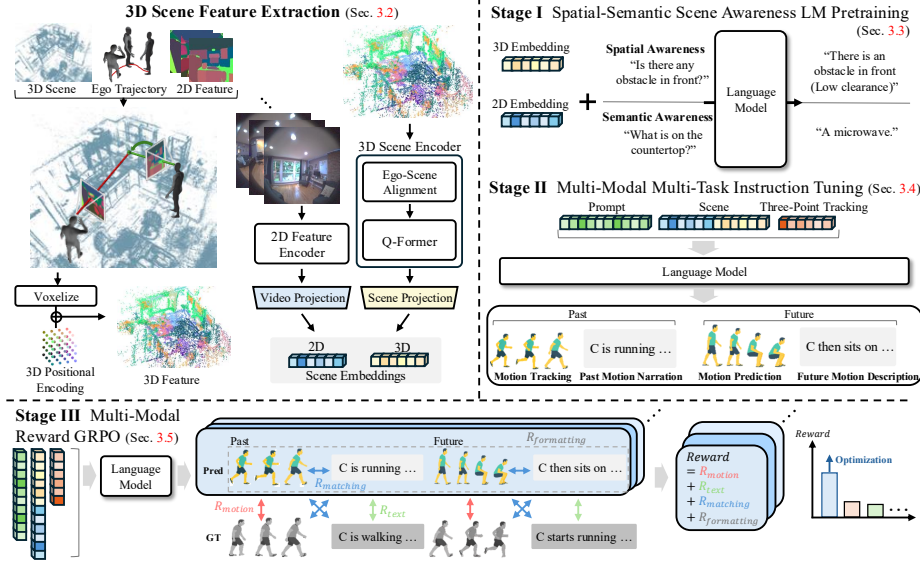


Fig. 2: Overview of the Ego3DLM framework. Our model consists of three stages. **3D Scene Feature Extraction** (Sec. 3.2): 2D semantic features are extracted from egocentric video frames and lifted onto the 3D point cloud. The feature-enhanced point cloud is fed into a Q-Former to produce compact scene query embeddings (top left). **LM Training: (I) Pre-training** (Sec. 3.3) aligns scene, motion, and language via two complementary objectives; **(II) Instruction Tuning** (Sec. 3.4) trains simultaneous single-pass generation of all four outputs, comprising past and future poses and descriptions (top right); **(III) GRPO** (Sec. 3.5) refines the model with intra- and inter-modal rewards for pose-language fidelity (bottom).

3 Method

We posit that predicting future human motion requires spatial and semantic understanding of the surrounding 3D environment, which defines the physical constraints and affordances of possible movements. Furthermore, human motion is inherently tied to the semantic interpretation of actions being performed, necessitating joint reasoning over motion and language rather than treating them independently. Grounded in this principle, we introduce Ego3DLM trained on three stages, as shown in Fig. 2. Spatial-Semantic Scene Awareness LM Pre-training (Stage I) encodes 3D scene context and injects it into the LM (Sec. 3.2) and trains spatial and semantic scene awareness (Sec. 3.3). Multi-Modal Multi-Task Instruction Tuning (Stage II) trains holistic motion tracking and prediction alongside past motion narration and future motion description (Sec. 3.4). Finally, Multi-Modal Reward GRPO (Stage III) boosts the accuracy and cross-modal coherence of generated pose and text (Sec. 3.5).

3.1 Preliminaries

Problem Formulation. Given three-point motion tracking features from 6D poses $\mathbf{P}$, egocentric video embeddings $\mathbf{V}$, and 3D scene embeddings $\mathbf{S}$, our Ego3DLM jointly predicts four outputs in a single autoregressive pass: past pose tokens x^{past} , future pose tokens x^{fut} , past motion narration y^{past} , and future motion description y^{fut} . Each pose sequence $x \in \{x^{\text{past}}, x^{\text{fut}}\}$ is represented as discrete motion tokens decoded into joint positions $J \in \mathbb{R}^{23 \times 3}$. Both past narration and future descriptions $y \in \{y^{\text{past}}, y^{\text{fut}}\}$ are natural language description of the corresponding motion segment. Our Ego3DLM produces a structured output sequence in a single autoregressive pass: $(x^{\text{past}}, x^{\text{fut}}, y^{\text{past}}, y^{\text{fut}}) = f_\theta(\mathbf{P}, \mathbf{V}, \mathbf{S})$, where f_θ is the language model.

Pose Representation and Tokenization. Following EgoLM [30], human motion is modeled as a 23-joint kinematic tree with 6D rotation parameterization. Pose of each frame t is represented as $M_t = [V_t^r, R_t^r, R_t^{rv}, R_t^j, R_t^{jv}] \in \mathbb{R}^{279}$, from which joint positions $J \in \mathbb{R}^{23 \times 3}$ are recovered via forward kinematics. To enable discrete generation, we train a PQ-VAE that maps pose sequences to indices over $C = 4096$ codebooks, extending the LM’s vocabulary with pose tokens $\{x\}_{c=1}^C$ and allowing the model to reason over pose and language in a unified token space. Further details are found in the supplementary materials.

Language Model. We build on a GPT-style causal LM (GPT-2, Qwen 2.5) [5, 59] trained primarily with next-token prediction. The LM comprises (i) a token embedding table (codebook), (ii) a Transformer [67] backbone, and (iii) an output projection layer that maps hidden states to vocabulary logits. Non-text inputs ($\mathbf{P}, \mathbf{V}, \mathbf{S}$) are injected as conditioning embeddings, enabling the model to exploit external context while preserving its linguistic priors. For motion tracking and forecasting, we use tokenized pose sequences, enabling the LM to generate motion outputs as discrete tokens.

3.2 3D Scene Feature Extraction

We design a pipeline that injects semantic information from egocentric video into the 3D scene point cloud. Following 3D-LLM [31], we generate dense frame-wise 2D semantic features using Mask2Former [11] instance masks and EVA-ViT-G [19] to fuse global and object-centric features. These are lifted onto the 3D point cloud using per-frame camera poses and a best-view selection policy, retaining features from the closest camera viewpoint for each 3D point to mitigate occlusion errors. The point cloud is transformed into the ego initial frame for alignment, voxelized for efficiency, and encoded with egocentric 3D sinusoidal positional encodings. A Q-Former [40] then compresses the scene into $K=32$ query embeddings. In parallel, per-frame CLIP [58] image embeddings are linearly projected into the LM embedding space. The video and 3D scene query embeddings together form the final scene conditioning for Ego3DLM.

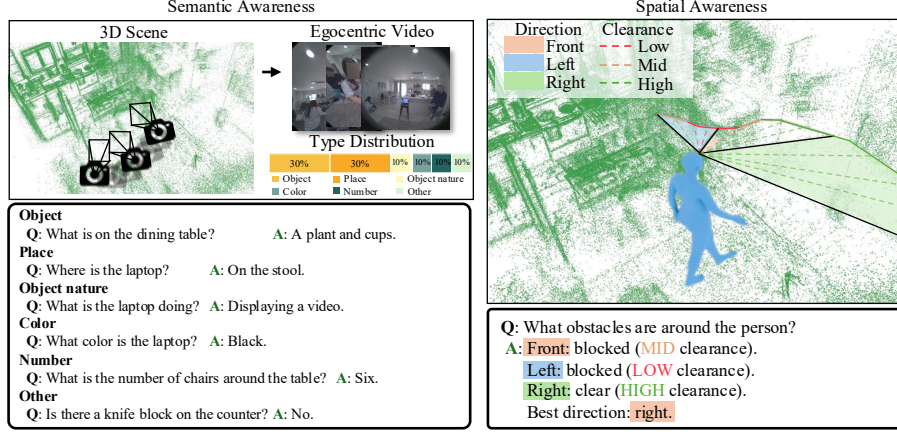


Fig. 3: Spatial-semantic scene awareness QA dataset generation. **Semantic Awareness:** For each 3D scene, we use egocentric videos to automatically generate question-answer pairs that describe and reason about the environment. We define six types of QA pairs targeting semantic understanding, and show their frequency distribution across the constructed data (left). **Spatial Awareness:** We partition the environment into three angular sectors (front, left, and right) and generate QA pairs to predict directional clearance levels and the most navigable direction (right).

3.3 Stage I: Spatial-Semantic Scene Awareness Pretraining

Human behavior is fundamentally constrained by environmental structure; therefore, understanding objects, their relations, and navigable free space is essential for downstream reasoning about human activity. In this light, we initialize the language model with strong scene understanding by training it to reason over the spatial layout and semantic attributes of the surrounding 3D environment. Given 3D scene embeddings $\mathbf{S}$ and egocentric video embeddings $\mathbf{V}$, we construct a dataset for spatial-semantic scene awareness by parsing the inputs into structured queries and answers as in Fig. 3. We then train the LM to jointly interpret semantic scene content (*e.g.* objects, attributes, and places) and spatial structure (*e.g.* obstacle layout and directional clearance). This stage equips the LM with a spatial-semantic prior over environments before introducing human motion reasoning in later stages.

Spatial Awareness (Q^{spa}). We partition the environment into three angular sectors (front, left, and right) and train the model to predict directional clearance levels (low, mid, high), obstruction state (blocked, clear), and the most navigable direction (front, left, right), encouraging the LM to internalize obstacle layouts from egocentric observations. **Semantic Scene Awareness (Q^{sem}).** We automatically generate scene-centric QA pairs capturing object identity, place relations, attributes, color, and counting using VLM [5], augmenting Nymeria [45] which lacks explicit scene descriptions. We filter samples relying on unreliably reconstructed structures and remove human-centric attributes.

Overall, the resulting dataset contains approximately 535K spatial QA pairs and 115K semantic QA pairs across 208 scenes. Further details on the dataset construction process for both are found in the supplementary materials. Given $\mathbf{E} = (\mathbf{S}, \mathbf{V})$ and a semantic query q , the LM predicts the answer sequence a :

$$\mathcal{L}_{\text{Pre}} = - \sum_i \left[\underbrace{\log p_\theta(a_i^{\text{sem}} \mid a_{<i}^{\text{sem}}, q^{\text{sem}}, \mathbf{E})}_{Q^{\text{sem}}} + \underbrace{\log p_\theta(a_i^{\text{spa}} \mid a_{<i}^{\text{spa}}, q^{\text{spa}}, \mathbf{E})}_{Q^{\text{spa}}} \right], \quad (1)$$

where semantic awareness queries and answers are denoted as $(q^{\text{sem}}, a^{\text{sem}})$ and spatial awareness queries and answers as $(q^{\text{spa}}, a^{\text{spa}})$, respectively. This encourages the LM to internalize both semantic scene knowledge and spatial layout reasoning. This spatial-semantic prior forms the foundation for subsequent stages that require reasoning about human behavior within the environment.

3.4 Stage II: Multi-Modal Multi-Task Instruction Tuning

We train the model using a unified instruction-tuning objective that jointly supervises four outputs: past motion, past motion description, future motion, and future motion description. Rather than predicting these outputs independently, the model generates them in a single autoregressive sequence, enabling shared reasoning across time and modalities. A key design choice is to prepend an explicit *spatial scene reasoning* step before any motion or language output. This encourages the model to first analyze spatial constraints in the environment, such as obstacles, free space, and navigable directions, before committing to predictions. The resulting generation process follows a chain-of-thought-like structure [29], where spatial reasoning informs motion prediction, and motion prediction subsequently informs semantic description.

Conditioned on 3D scene embeddings $\mathbf{S}$, egocentric video embeddings $\mathbf{V}$, and three-point tracking features $\mathbf{P}$, the model produces outputs in the following structured order: [spatial scene description] $\rightarrow$ [past & future motion] $\rightarrow$ [past & future motion description]. By generating a spatial scene description, motions, and motion descriptions in a single autoregressive pass, the model propagates structured context across the full output sequence, promoting three complementary forms of reasoning: *spatial understanding*, where the model infers scene constraints to guide subsequent motion prediction; *temporal understanding*, where past and future motion are jointly modeled so that observed trajectories directly inform future prediction; and *semantic grounding*, where simultaneously predicting motions and descriptions enforces that the physical realization of a motion and its semantic description remain mutually consistent. Given input $(\mathbf{S}, \mathbf{V}, \mathbf{P})$ and a spatial query q^{spa} , we construct an instruction prompt $\mathbf{I}$, as detailed in the supplementary materials. Given targets $(a^{\text{spa}}, x^{\text{past}}, x^{\text{fut}}, y^{\text{past}}, y^{\text{fut}})$, we optimize a unified objective that jointly super-

vises all output components:

$$\begin{aligned}\mathcal{L}_{\text{IT}} = & \log p_{\theta}(a_i^{\text{spa}} | a_{<i}^{\text{spa}}, \mathbf{I}) + \log p_{\theta}(x_i^{\text{past}} | x_{<i}^{\text{past}}, a^{\text{spa}}, \mathbf{I}) \\ & + \log p_{\theta}(x_i^{\text{fut}} | x_{<i}^{\text{fut}}, x^{\text{past}}, a^{\text{spa}}, \mathbf{I}) + \log p_{\theta}(y_i^{\text{past}} | y_{<i}^{\text{past}}, x^{\text{fut}}, x^{\text{past}}, a^{\text{spa}}, \mathbf{I}) \\ & + \log p_{\theta}(y_i^{\text{fut}} | y_{<i}^{\text{fut}}, y^{\text{past}}, x^{\text{fut}}, x^{\text{past}}, a^{\text{spa}}, \mathbf{I})\end{aligned}\quad (2)$$

3.5 Stage III: Multi-Modal Reward GRPO

Supervised instruction tuning optimizes the likelihood of ground-truth outputs but does not directly encourage cross-modal coherence between simultaneously generated motions and descriptions. To address this, we introduce a reinforcement learning stage using Group Relative Policy Optimization (GRPO) [63], which optimizes the model toward outputs that are simultaneously accurate and internally consistent across modalities.

GRPO Objective. For each prompt $\mathbf{I}$, GRPO samples a group of G candidate outputs $\{o_i\}_{i=1}^G$ from the old policy π_{old} , assigns each a scalar reward r_i , and updates the policy by maximizing:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\rho_{i,t} \hat{A}_{i,t}, \text{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}] \right\}, \quad (3)$$

where $\rho_{i,t} = \pi_{\theta}(o_{i,t} | o_{i,<t}, \mathbf{I}) / \pi_{\theta_{\text{ref}}}(o_{i,t} | o_{i,<t}, \mathbf{I})$ is the importance ratio and the group-relative advantage is:

$$\hat{A}_{i,t} = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}, \quad \mathbf{r} = \{r_1, \dots, r_G\}. \quad (4)$$

This eliminates the need for a separate value network, as outputs scoring above the group mean are reinforced purely through within-group reward comparisons.

Multi-Modal Reward. The scalar reward r_i decomposes into intra-modal and inter-modal signals, applied symmetrically over both past and future outputs:

$$r = w_m \cdot \underbrace{\max(0, 1 - \text{JPE})}_{R_{\text{motion}}} + w_t \cdot \underbrace{\text{BLEU}}_{R_{\text{text}}} + w_d \cdot R_{\text{matching}} + R_{\text{format}}. \quad (5)$$

JPE (global Joint Position Error) and BLEU-4 are intra-modal rewards that independently measure motion accuracy and description quality against ground truth. R_{format} penalizes outputs that violate the structured output template. w is the weighting constant between rewards.

The inter-modal reward R_{matching} is the key signal that supervised training cannot provide, directly measuring fidelity between the simultaneously predicted motion and description:

$$R_{\text{matching}} = - \left[d_{gp}(\mathbf{e}_{t_{\text{gt}}}, \mathbf{e}_{m_{\text{pred}}}) + d_{pg}(\mathbf{e}_{t_{\text{pred}}}, \mathbf{e}_{m_{\text{gt}}}) + d_{pp}(\mathbf{e}_{t_{\text{pred}}}, \mathbf{e}_{m_{\text{pred}}}) \right], \quad (6)$$

where $d(\cdot, \cdot)$ is Euclidean distance in a shared embedding space projected by modality-specific encoders [24], and $\mathbf{e}_t, \mathbf{e}_m$ denote text and motion embeddings respectively. Each term penalizes inconsistency between: (i) ground-truth description and predicted motion (d_{gp}), (ii) predicted description and ground-truth motion (d_{pg}), and (iii) the simultaneously predicted motion-description pair (d_{pp}). Together, they encourage the model to produce motion and description that are mutually grounded, a signal that likelihood-based training alone cannot provide.

4 Experiments

4.1 Experimental Setup

Dataset. We conduct experiments on the Nymeria dataset [45], a large-scale in-the-wild egocentric human motion corpus recorded with Project Aria devices [16], containing language narrations, egocentric video, and 3D point clouds. We parse each recording into narrated motion segments and form past-future pairs by linking successive 5-second segments at 10 FPS. The earlier segment serves as past context, and the subsequent segment as the future target. We split the data at the scene level so that training and validation sets do not share environments, and we stratify scenes by dominant motion pattern (largely static vs. dynamic human motion) to maintain a similar static/dynamic ratio in both splits. Further details are found in the supplementary materials.

Baselines. We compare ours with 3 baseline methods: EgoLM [30], FIction [1], and UniEgoMotion [52], all of which predict future pose sequences. EgoLM separately generates text narration for past motion tracking along with human poses. Other baselines only generate motion without language. All baseline methods utilize three-point tracking and egocentric video, and FIction and UniEgoMotion additionally utilize a 3D scene. Results for EgoLM with a 3D scene can be found in the supplementary materials.

Metrics. APE, JPE, and ADE are pose-related metrics that respectively measure the local joint pose accuracy, global joint pose accuracy, and global head pose accuracy. $2s$ denotes the accuracy of the first 2 seconds. For tracking we additionally use Upper and Lower for upper body and lower body APE, along with joint angle (J. A.) and Root. Joint angle (J.A.) measures the mean angular error across all body joints, and Root measures the positional error of the root joint which captures global translation accuracy. For text, we use Bleu [50], Rouge_L [41], SBert score [61], and R-precision. $R@N$ denotes top-N R-precision accuracy of narration. Motion-description matching distance metrics d_{gp} , d_{pg} , and d_{pp} are defined as L2 distances in the embedding space produced by modality specific encoders [24], computed between ground truth description and predicted motion (d_{gp}), predicted description and ground truth motion (d_{pg}), and the predicted motion-description pair (d_{pp}), respectively. d_{pp}^f and d_{pp}^p denote the motion-description matching distances for the future and past. For pretraining on $\mathcal{Q}^{spa}$, collision (obstruction) and freespace metrics measure the mean accuracy of all directions. Further details are found in the supplementary materials.

Table 1: Comparison of prediction and tracking results on motion and text metrics. x - y Align. denotes motion and text alignment. Embedding distance functions (d_{gp} , d_{pg} , d_{pp}) are defined in Sec. 3.5. Best are denoted in **bold**. Arrow denotes better.

Method	Motion Prediction (3 modes)								Motion Tracking				
	APE ↓	JPE ↓	ADE ↓	ADE _{2s} ↓	FDE ↓	FDE _{2s} ↓	FID ↓	Diversity ↑	APE ↓	Upper ↓	Lower ↓	J.A. ↓	Root ↓
Fiction [1]	206.2	564.7	558.3	416.7	904.3	494.9	0.3275	0.7432	181.1	114.0	282.3	31.28	34.81
EgoLM [30] (GT motion)	168.8	583.9	540.0	299.0	1,059.5	494.8	1.3840	0.0055	-	-	-	-	-
EgoLM [30] (Inst. tuning)	184.9	579.4	552.6	329.9	983.6	519.2	0.2137	0.6142	161.9	95.6	265.6	33.63	24.01
UniEgoMotion [52]	151.5	424.3	409.7	223.9	720.5	360.4	0.1530	0.9022	152.2	79.4	233.3	26.48	22.71
Ours (Ego3DLM)	147.9	364.5	343.9	205.9	648.1	312.6	0.0160	1.0624	96.4	53.1	152.7	22.30	19.57

Method	Future Motion Description						Past Motion Narration						x - y Align.
	Bleu-4 ↑	Bleu-1 ↑	Rogue _L ↑	SBert ↑	R@3 ↑	d_{pp} ↓	Bleu-4 ↑	Bleu-1 ↑	Rogue _L ↑	SBert ↑	R@3 ↑	d_{pp} ↓	d_{pp} ↓
LLM (Qwen 2.5 7B [5])	0.0202	0.0970	0.2343	0.6120	0.1940	6.6730	0.0232	0.1477	0.2267	0.6183	0.1944	6.9648	-
EgoLM [30] (Inst. tuning)	-	-	-	-	-	-	0.0649	0.3002	0.2642	0.5900	0.2639	6.8992	9.8686
Ours (Ego3DLM)	0.1039	0.3839	0.3004	0.6206	0.2911	6.3507	0.1107	0.3966	0.3195	0.6458	0.4264	4.9470	4.2571

Implementation Details. We use a GPT-2 medium as the language model backbone. Full training settings, additional details, including pretraining and instruction templates, 3D scene QA dataset generation, and additional results with Qwen 2.5 backbone, are provided in the supplementary materials.

4.2 Quantitative Prediction

As shown in Table 1, Ego3DLM achieves state-of-the-art performance across all motion prediction and tracking metrics. For prediction, our model outperforms the strongest baseline UniEgoMotion across all metrics, reducing APE by 2.4%, JPE by 14.1%, ADE_{2s} by 8.0%, and FDE_{2s} by 13.3%, while also achieving the best FID and highest diversity, indicating that our predictions are both accurate and diverse. The improvement is even more pronounced for motion tracking, where Ego3DLM reduces APE by 36.7% and upper/lower body APE by 33.1%/34.5% over UniEgoMotion, demonstrating that joint past-and-future reasoning with explicit 3D scene conditioning yields a substantially more accurate understanding of observed motion.

One aspect to note is that Ego3DLM’s holistic prediction is superior to modality-specific or separately generated outputs. Methods that predict only motion (Fiction, UniEgoMotion) forgo language entirely, while a general-purpose LLM (Qwen 2.5 7B) produces text without any motion grounding, and EgoLM, despite generating both modalities, does so through separate decoding streams. In contrast, our single-pass joint decoding achieves substantially higher scores across all text metrics, improving future motion description Bleu-4 by 60.0% and past motion narration Bleu-4 by 70.4% over EgoLM. Most strikingly, the motion-description embedding distance (x - y Align.) improves by 56.8%, indicating that our simultaneously predicted motion and descriptions are tightly aligned in a shared semantic space. This demonstrates that Ego3DLM produces physically plausible motions whose simultaneously generated motion descriptions faithfully reflect the underlying motion dynamics, confirming that holistic joint prediction yields semantically coherent motion-language outputs.

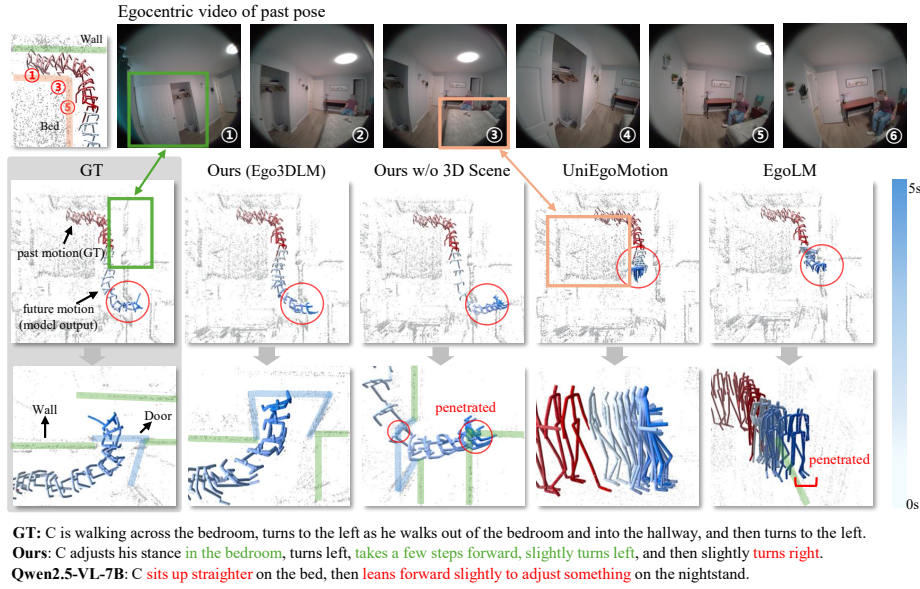


Fig. 4: Qualitative results for motion forecasting and tracking. The ground-truth (GT) past motion trajectory is shown in **red**, and the predicted future motion is visualized in **blue**. For future motion narration below the figure, **green** highlights correct phrases matching the GT, while **red** highlights incorrect or mismatched descriptions.

4.3 Qualitative Prediction

Figure 4 presents qualitative comparisons among UniEgoMotion, EgoLM, and our Ego3DLM with and without 3D scene. For motion forecasting, ours without 3D scene fails to incorporate the room layout and collides with a wall. While UniEgoMotion leverages 3D scene to avoid collision, it generates a contextually irrelevant motion that abruptly stops. As EgoLM does not leverage 3D scene, the predicted motion results in collision. Our method generates an accurate sequence that predicts walking motion around the bed to the door. Notably, the future motion description accurately depicts the predicted motion, demonstrating a comprehensive knowledge of human motion intent in both motion and language domain. Overall, our model produces motions that best conform to both physical constraints and past motion context, outperforming baselines with or without 3D spatial context.

4.4 Ablation Study

Pretraining. Table 2 ablates the effect of input modality and pretraining target on scene understanding. First, combining both video and 3D scene inputs consistently improves performance over video alone across most metrics for both semantic awareness Q^{sem} and spatial awareness Q^{spa} , confirming that 3D

Table 2: Ablation on pretraining. The effects of input modality (video, 3D scene) and pretraining objective (Q^{sem} , Q^{spa}) are analyzed.

Objective	Video	3D Scene	Bleu-1 $\uparrow$	Bleu-4 $\uparrow$	SBert $\uparrow$	Rouge $_L$ $\uparrow$	Collision $_{mean}\uparrow$	Freespace $_{mean}\uparrow$
Q^{sem}	$\checkmark$		0.5298	0.2116	0.7403	0.5244	-	-
	$\checkmark$	$\checkmark$	0.5330	0.2478	0.7393	0.5219	-	-
Q^{spa}	$\checkmark$		-	-	-	-	0.7387	0.8720
	$\checkmark$	$\checkmark$	-	-	-	-	0.7428	0.8728
$Q^{sem} + Q^{spa}$	$\checkmark$	$\checkmark$	0.5399	0.2469	0.7438	0.5278	0.7764	0.8848

Table 3: Ablation on instruction tuning. The effects of pretraining (Q^{sem} , Q^{spa}), use of spatial scene reasoning (SSR) and 3D scene (**S**) are compared.

Method	Motion Prediction (3 modes)						Future Motion Description			
	APE $\downarrow$	JPE $\downarrow$	ADE $_{2s}$ $\downarrow$	FDE $_{2s}$ $\downarrow$	FID $\downarrow$	Diversity	R@3 $\uparrow$	R@2 $\uparrow$	R@1 $\uparrow$	d_{pg} $\downarrow$
Ours SFT only	148.5	368.3	208.8	317.9	0.0190	0.9358	0.2706	0.1903	0.1072	6.4837
w/o Q^{sem}	153.5	381.1	221.9	335.9	0.0336	0.9644	0.2558	0.1760	0.0987	6.5460
w/o Q^{spa}	152.9	378.3	214.3	329.5	0.0335	1.0122	0.2453	0.1725	0.0894	6.7133
w/o SSR	151.5	376.6	213.4	325.3	0.0231	1.0830	0.2638	0.1859	0.1021	6.4494
w/o S	156.2	388.5	220.8	334.3	0.0426	0.9352	0.2574	0.1809	0.0945	6.6789

Method	Motion Tracking					Past Motion Narration			x - y Align.	
	APE $\downarrow$	Upper $\downarrow$	Lower $\downarrow$	J.A. $\downarrow$	Root $\downarrow$	R@3 $\uparrow$	R@2 $\uparrow$	R@1 $\uparrow$	d_{pg} $\downarrow$	d_{pp} $\downarrow$
Ours SFT only	98.3	53.4	155.3	22.52	19.62	0.4157	0.3022	0.1693	5.0049	5.9170
w/o Q^{sem}	122.1	69.6	190.1	27.84	25.40	0.3960	0.2961	0.1674	5.1349	5.9190
w/o Q^{spa}	99.3	53.6	158.7	22.79	19.93	0.3987	0.2971	0.1697	5.2278	6.1612
w/o SSR	99.4	53.1	159.8	22.64	19.25	0.4090	0.3069	0.1661	5.0103	6.3930
w/o S	105.8	57.5	168.8	24.03	20.92	0.3955	0.2837	0.1566	5.2103	6.3012

scene features provide complementary spatial information beyond what egocentric video alone can convey. Second, jointly training on both spatial and semantic QA objectives yields the best results across all metrics, with Bleu-1, SBert, and Rouge $_L$ all reaching their peak while simultaneously achieving the highest Collision and Freespace scores. Notably, adding spatial awareness improves semantic understanding beyond training on semantic awareness alone, and vice versa, suggesting that the two objectives are mutually reinforcing rather than competing. This indicates that reasoning about spatial layout and semantic scene content are complementary skills, and that exposure to both during pretraining allows Ego3DLM to acquire a thorough understanding of the 3D environment.

Instruction Tuning. Table 3 ablates the contribution of each component during instruction tuning across all four tasks. Removing 3D scene input causes the most consistent and severe degradation across all tasks, confirming that explicit 3D scene conditioning is the single most important factor for accurate motion prediction, as the model cannot adequately infer physical constraints from egocentric video alone. Removing semantic QA pretraining leads to a sub-

Table 4: Ablation on GRPO finetuning. A more comprehensive reward signal results in overall improvement, especially the semantic understanding of past and future motion. d_{pp}^f and d_{pp}^p each denote the embedding distance between generated motion and description for future and past, respectively.

Reward			Motion Prediction (3 modes)					Motion Tracking				
R_{motion}	R_{text}	R_{matching}	APE ↓	JPE ↓	ADE _{2s} ↓	FDE _{2s} ↓	FID	APE ↓	Upper ↓	Lower ↓	J.A. ↓	Root ↓
SFT only			148.5	368.3	208.8	317.9	0.0190	98.3	53.4	155.3	22.52	19.62
✓			148.1	364.2	204.8	310.5	0.0149	96.6	53.1	153.2	22.43	19.80
✓	✓		148.7	369.6	206.2	313.8	0.0177	97.4	53.4	153.6	22.55	19.60
✓	✓	✓	147.9	364.5	205.9	312.6	0.0160	96.4	53.1	152.7	22.30	19.57

Reward			Future Motion Description				Past Motion Narration				x - y Align.	
R_{motion}	R_{text}	R_{matching}	Bleu-4 ↑	Bleu-1 ↑	R@3 ↑	d_{pg} ↓	Bleu-4 ↑	Bleu-1 ↑	R@3 ↑	d_{pg} ↓	d_{pp}^f ↓	d_{pp}^p ↓
SFT only			0.1036	0.3789	0.2706	6.4837	0.1126	0.3930	0.4157	5.0049	5.9170	4.2861
✓			0.1019	0.3716	0.2792	6.3974	0.1148	0.3936	0.4206	4.9424	5.9181	4.3260
✓	✓		0.1037	0.3815	0.2726	6.4560	0.1160	0.3944	0.4071	5.0080	5.8519	4.3583
✓	✓	✓	0.1039	0.3839	0.2911	6.3507	0.1107	0.3966	0.4264	4.9470	5.8063	4.2571

stantial drop in motion tracking performance, suggesting that semantic scene understanding provides crucial context for interpreting observed body motion relative to the environment. Removing spatial QA similarly degrades both motion prediction and cross-modal alignment, indicating that spatial awareness of navigable directions and clearance is particularly important for generating physically plausible future motion and grounding them in language. Together, these results corroborate the pretraining ablation findings: semantic and spatial understanding are complementary, and both contribute to downstream prediction quality. Finally, removing spatial scene reasoning (SSR) leads to notable degradation in motion-description alignment and future motion description quality. This demonstrates that the structured scene reasoning step does not merely reformat outputs; it actively elicits spatially grounded reasoning that propagates through the generation process, improving the semantic coherence between predicted motions and their descriptions.

GRPO Finetuning. Table 4 validates the contribution of each reward component. As expected, adding R_{motion} primarily improves motion prediction (JPE by 1.1%, FID by 21.6%), and further adding R_{text} improves narration quality (Bleu-4 by 3.0%), confirming that intra-modal rewards effectively sharpen their respective output modalities. Notably, incorporating the inter-modal matching reward R_{matching} yields broad improvements across *both* motion and text metrics simultaneously, including motion prediction, narration Bleu scores, and motion-description alignment distances (d_{pp}^f by 0.8%, d_{pp}^p by 2.3% over the previous stage). This reciprocal improvement demonstrates that explicitly rewarding motion-description consistency encourages the model to develop a comprehensive and semantically grounded understanding of human motion intent. By enforcing

predicted motions and descriptions to be mutually coherent, the model learns richer cross-modal representations that benefit each modality independently.

5 Conclusion

We present Ego3DLM, a unified 3D-aware framework for holistic tracking and forecasting of all four outputs in a single autoregressive pass: past motion, future motion, past narration, and future description. By leveraging three-point tracking, egocentric video, and explicit 3D scene features, Ego3DLM establishes a new state-of-the-art across all motion prediction, tracking, and language description benchmarks. This is achieved by spatial-semantic scene awareness pretraining, holistic multi-task instruction tuning, and multi-modal reward GRPO finetuning that directly optimizes motion-language fidelity. Our experiments demonstrate consistent improvements over baselines, particularly in the tight coupling of predicted motion and its description, marking a step toward more anticipatory, interpretable, and interactive embodied AI systems.

Limitations and Future Works. Since most human activities take place in familiar venues where users spend substantial time and accumulate rich 3D spatial knowledge, we assume the 3D scene feature is given. While reasonable for most XR applications, reliance on a precomputed 3D scene remains a limitation. This opens up promising future work toward a language model that can incrementally register 3D knowledge of newly observed scenes on the fly and reason about how a human will interact within them.

Acknowledgements

This work was supported by the Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No. RS-2024-00457882, AI Research Hub Project), by Center for Advanced Urban Systems (CAUS) of Korea Advanced Institute of Science and Technology (KAIST) funded by GS E&C, and by the InnoCORE program of the Ministry of Science and ICT(N10250156).

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